

BTCS501**VI Semester Examination, May – June 2022****B.Tech. (CCE/CSE/CSE-AII/CSE-ES/CSE-ICS/CSE-BDAI/CSE-MA/CSE-CMCI/IT)****B.Tech.+ M.Tech. (CSE/CSE-CC/CSE-CF/CSE-BDA/IT)****B.Tech.+ M.B.A. (CSE/IT)****THEORY OF COMPUTATION**

Choice Based Credit System (CBCS)

Time: 3 Hrs.**Maximum Marks: 60****Minimum Pass Marks: 24**

Note: All questions carry equal marks, out of which part 'A' and 'B' carry 3 marks and part 'C' carries 6 marks.
 From each question, part 'A' and 'B' are compulsory and part 'C' has internal choice.
 Draw neat diagram, wherever necessary.
 Assume suitable data wherever necessary.

Q.1(A) Explain the following with examples:

1. Automata

2. DFA

3. NFA

03

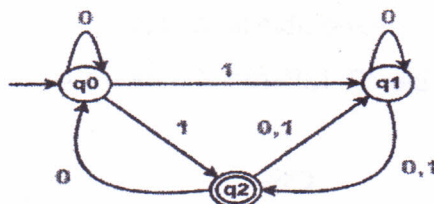
(B) Construct a Finite Automata accepting all string over $\{0,1\}$ having even number of 0's and even number of 1's.

03

(C) State and prove Myhill - Nerode Theorem.

OR

(D) Convert the following Non-Deterministic Finite Automata (NFA) to Deterministic Finite Automata (DFA)

**06**

Q.2(A) State and Define the closure properties of regular languages.

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(B) What is Mealy Machine? How Finite Automates can be converted into Moore Machine? Explain with the help of example.

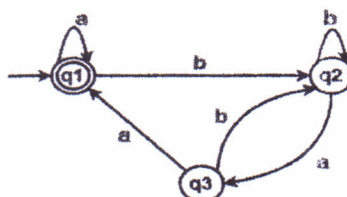
03

(C) Explain pumping lemma for regular language. Prove following language is not regular.

$$L = \{ a^n : n \geq 1 \}$$

OR

(D) Find out the Regular Expression from given DFA

**06****Contd...**

Q.3(A) Explain Chomsky Normal Form with example.

03

(B) Give Context Free Grammar for Regular Expression $(011+1)^*(01)^*$

03

(C) Check whether the following grammar is ambiguous or not. Remove the ambiguity from the grammar in case the grammar is ambiguous.

$S \rightarrow abAB$
 $A \rightarrow bAB \mid \lambda$
 $B \rightarrow BAa \mid A \mid \lambda$

OR

(D) Convert the grammar into Greibach Normal Form.

$S \rightarrow ABb \mid a$
 $A \rightarrow aaA \mid B$
 $B \rightarrow bAb$

06

Q.4(A) Compare Pushdown Automata with context free grammar.

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(B) Convert the following Context Free Grammar to Pushdown Automata.

$S \rightarrow aAA$
 $A \rightarrow As \mid bS \mid a$

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(C) Construct pushdown automata for the following languages. Accept it either by empty stack or by final state. $\{a^n b^m a^n : m, n \in \mathbb{N}\}$

OR

(D) Construct the PDA for the following example $\{a^m b^n c^{m+n} : m, n \geq 1\}$

06

Q.5(A) What are decidable and undecidable problems?

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(B) Give two properties of recursively enumerable language.

03

(C) What is meant by halting problem? Differentiate between P, NP hard and NP complete problems.

OR

(D) Explain Turing machine model. Design a Turing Machine for the language

$\{L(G) = a^n b^n : n \geq 1\}$

06
